

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (BT/MI/BC) Examination April-May 2019

Course code & Name: BS601 Bioinformatics

Date: 30/04/2019 Day: Tuesday Time: 1:30 PM To 1:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. PRINTS are used for _____.
(i) detection of genes from genome (ii) detection of tRNA genes
(iii) prediction the function of new gene (iv) identification of functional domains/ motifs of protein
2. Which is data retriving tool?
(i) ENTREZ (ii) EMBL
(iii) BLAST (iv) All of these
3. Which of the following correctly represent protein structure classification path in SCOP?
(i) Class → Fold → Super family → Family → Protein
(ii) Fold → Class → Family → Super family → Protein
(iii) Protein → Fold → Super family → Family → Class
(iv) Protein → Class → Fold → Family → Super family
4. BLAST X program is used for _____.
(i) translated protein sequence (ii) translated input sequence
(iii) translated DNA database (iv) none of these

5. Which secondary structure is formed when amino acid residues have negative value of ϕ (phi) angle and negative value of ψ (psi) angle?
- (i) Antiparallel β sheet (ii) Parallel β sheet
- (iii) Right handed alpha helix (iv) none of these
6. Which of the following tool used for protein structural alignment?
- (i) VAST (ii) RASMOL
- (iii) DALI (iv) (i) and (iii) both
7. Which of the following is application of MSA?
- (i) Domain identification (ii) Phylogenetic analysis
- (iii) PCR analysis (iv) All of these
8. In which sequence format first line of each sequence entry begins with a "greater than", (>) sign and is ended with an "asterisk" (*)?
- (i) FASTA (ii) NBRF
- (iii) Genbank (iv) GCG
9. Phylogram show common ancestry, but do not indicate the amount of evolutionary "time" separating taxa.
- ☐ True ☐ False
10. The MMDB, PDB and EMBL databases share data on a daily basis and are therefore equivalent.
- ☐ True ☐ False

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Course code & Name: BS601 Bioinformatics

Date: 30/04/2019 Day: Tuesday Time: 1:50 pm To 3:30 pm Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q – II Answer the following questions as directed (any five)

10

1. Explain in brief about Genbank format.
2. Define: a) Indel b) Orthologous gene
3. Define: a) Bit score b) Query coverage
4. Write the differences between global and local alignment.
5. Write a note on “scope of bioinformatics”.
6. What is data mining? Draw the flow of data mining process.

SECTION – II

Q-III Answer the following questions as directed (any five)

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1. What is MSA? How MSA can be used for universal primer designing.
2. What is Data database? Briefly describe secondary protein database.
3. What is phylogenic tree? Explain it with appropriate example.
4. Write a note on “Molecular Modeling Database”
5. What is bioinformatics? Explain any two application of bioinformatics with example.
6. Identify and analyze the below mentioned BLAST output. Give your comment on it.

```
Score = 790 bits (2040), Expect = 0.0
Identities = 520/1381 (37%), Positives = 745/1381 (53%), Gaps = 36/1381 (2%)
Frame = +3/+3
Query: 156 SEMNVNMKYQLPNFTAETPIQNVVLHKHH--IYLGA VNYIYVLNDKDLQKVAEYKTGPVL 329
      S +N ++ Y +P F A PIQN+V + + +Y+ + N I +N + L+KV E +TGPV
Sbjct: 31 SPVNFSVVYTMPFFQAGGPIQNI VNN SFYQEVYVASQNVIEAVN-QSLEKVVWELRTGPV- 88
```

7. Analyze the following Dot Matrix and write down the two aligned sequence.
(indicate indels and gaps in aligned sequence)

